

Avancini Lara

Senior AI Engineer | LLM Systems, AI Platforms & Backend Engineering

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Professional Summary

Senior AI and software engineer with experience building production LLM systems, data platforms, automation, and backend services. Strong background in Python, FastAPI, Apache Airflow, SQL/PostgreSQL, cloud infrastructure, and end-to-end AI application development. Experienced in taking systems from architecture through production, including LLM evaluation and optimization, structured extraction and classification, workflow orchestration, internal tooling, and reliability improvements. Background in computational physics and scientific programming.

Experience

TELUS Digital | Senior AI Engineer

Jul 2026 - Present | Remote

- Design and build an LLM-assisted platform for identifying, evaluating, and ranking high-quality customer-support interactions across datasets containing hundreds of thousands of transcripts per month, including an automated feedback loop that validates LLM-generated SQL against user requirements and supports iterative corrections through natural-language feedback.
- Designed a Next.js interface and FastAPI backend for reviewing, explaining, editing, and executing generated queries, replacing an initially model-specific prototype with an extensible, OpenAI-compatible API.
- Expanded the platform with eligibility checks, quality scoring, and clustering workflows to further evaluate, filter, and organize candidate conversations.
- Reduced LLM inference costs by more than 90% and made per-transcript processing approximately 3x faster by benchmarking models across accuracy, latency, and cost and moving workloads to more efficient domain-appropriate models.
- Deployed and currently operate the application on Google Cloud Platform using Cloud Run, Cloud Build, Secret Manager, and OAuth 2.0, with ownership spanning frontend, API, LLM workflows, and production deployment.

Telescope Partners | Senior Software Engineer

Jul 2025 - Jun 2026 | USA (Remote)

- Owned and evolved production workflows built with Apache Airflow, improving reliability, observability, and day-to-day operation of data and automation pipelines.
- Designed and implemented LLM-powered extraction and classification pipelines using structured outputs, prompt engineering, validation, and failure handling for real-world inputs.
- Improved LLM cost control through prompt consolidation, caching, token limits, and safeguards against runaway consumption.
- Migrated data workloads from Tinybird-style implementations to PostgreSQL-backed architectures, focusing on query performance, maintainability, and operational simplicity.
- Built reusable Python libraries and internal tooling that standardized workflow patterns and reduced duplicated implementation work across projects.

Thoughtful AI | Senior Software Engineer

Jul 2024 - Jul 2025 | USA (Remote)

- Designed reusable internal Python libraries used across multiple projects, reducing duplicated implementation work and standardizing common development patterns.
- Designed architecture for automation and backend systems with an emphasis on maintainability, performance, and production reliability.
- Built and maintained healthcare automation products integrating internal systems, third-party applications, and external APIs.

Thoughtful AI | Software Engineer

Jun 2023 - Jul 2024 | USA (Remote)

- Developed Python automation for healthcare workflows using RPAFramework, Selenium, browser automation, and reverse-engineered APIs.
- Integrated healthcare systems without conventional APIs by analyzing application network traffic and reproducing required requests programmatically.
- Automated workflows involving patient, financial, and operational data while maintaining data accuracy and handling production failure cases.
- Monitored and improved deployed automations based on runtime behavior, errors, performance, and changes in external systems, collaborating with engineering and operations teams to diagnose issues.

KarHub | Python Developer, Data Team

2022 - Jun 2023 | São Paulo, Brazil (Remote)

- Developed ETL pipelines for collecting, transforming, and processing business data using Python and Pandas.
- Built web scrapers, data collection systems, and RPA workflows using Selenium and other Python tooling.
- Developed REST APIs with Flask and FastAPI and used RabbitMQ for asynchronous processing and service communication.
- Built serverless workloads with AWS Lambda and used Apache Airflow for scheduling, orchestration, and monitoring of data pipelines.

UNESP - Nonlinear Dynamics Laboratory | Nonlinear Dynamics Researcher, CNPq Scholarship

2017 - 2020 | Rio Claro, Brazil

- Developed computational models for nonlinear dynamical systems and translated an existing FORTRAN model to Python.
- Performed numerical and statistical analysis of chaotic systems, including the Dissipative Standard Map, using Python, C, and FORTRAN.
- Created scientific visualizations with Matplotlib and Gnuplot for analysis and presentation of research results.

Technical Skills

Programming: Python, SQL

AI & LLM Engineering: LLM application development, evaluation and benchmarking, structured outputs, information extraction, classification, prompt engineering, model selection and routing, feedback loops, inference optimization, OpenAI-compatible APIs

AI Platforms & Models: Anthropic Claude, Google Vertex AI, open-weight and self-hosted LLMs

Backend & Frontend: FastAPI, Flask, REST APIs, Next.js, React, PostgreSQL

Data & Automation: Apache Airflow, Pandas, ETL, RPAFramework, Selenium, RabbitMQ, web scraping

Cloud & Infrastructure: Google Cloud Platform (Cloud Run, Cloud Build, Secret Manager, OAuth 2.0), AWS Lambda, Amazon S3, Docker

Education

Federal University of Rio Grande do Sul (UFRGS)

Bachelor's Degree in Physics

2020 - 2022 | Porto Alegre, Brazil

São Paulo State University (UNESP)

Physics - transferred before completion

2017 - 2020 | Rio Claro, Brazil